SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 **Total Marks:40**

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS)**.** Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer**.** Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer**.** Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT)**.** Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

2. **Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.**

Aim

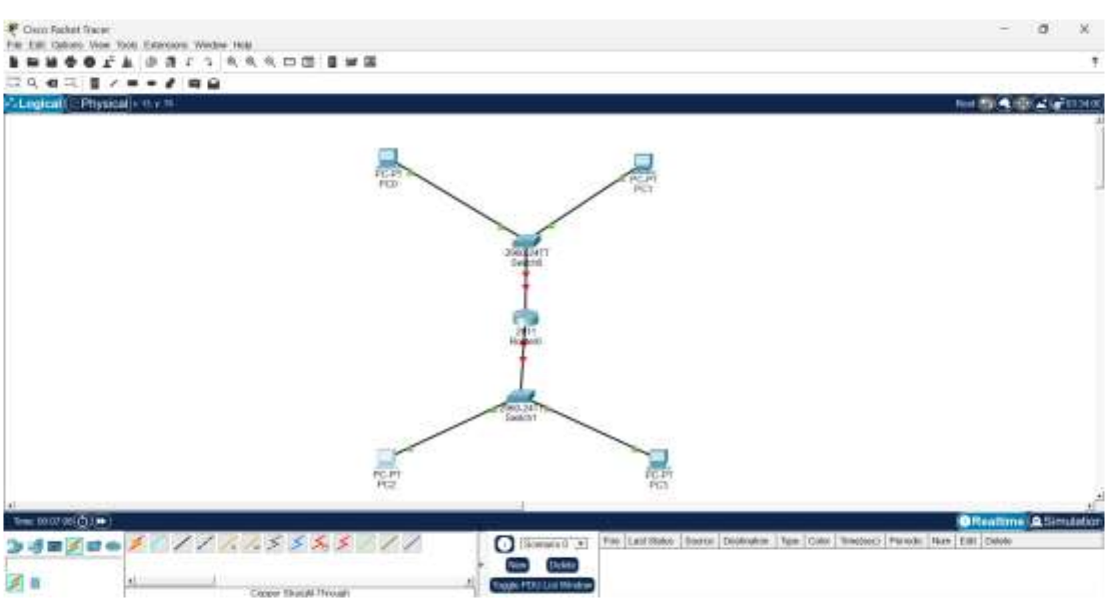
To design and configure two LANs connected through a router in Cisco Packet Tracer, verify end-to-end communication using ping, and analyze IPv4/ICMP packet header fields using Wireshark.

Procedure

1. Create two LANs using PCs and switches, and connect both LANs to a router.
2. Assign suitable IPv4 addresses and subnet masks to all PCs and router interfaces.
3. Configure the default gateway of each PC according to its respective LAN.
4. Enable the router interfaces using the no shutdown command.
5. Verify connectivity within and between the two LANs using the ping command.
6. Start Wireshark and capture the ICMP packets generated during the ping operation.
7. Analyze the IPv4 header fields such as Source IP, Destination IP, TTL, Protocol Number, and Header Length.
8. Observe how the router forwards packets between different networks through the Network layer.

Implementation

Cisco Packet Tracer



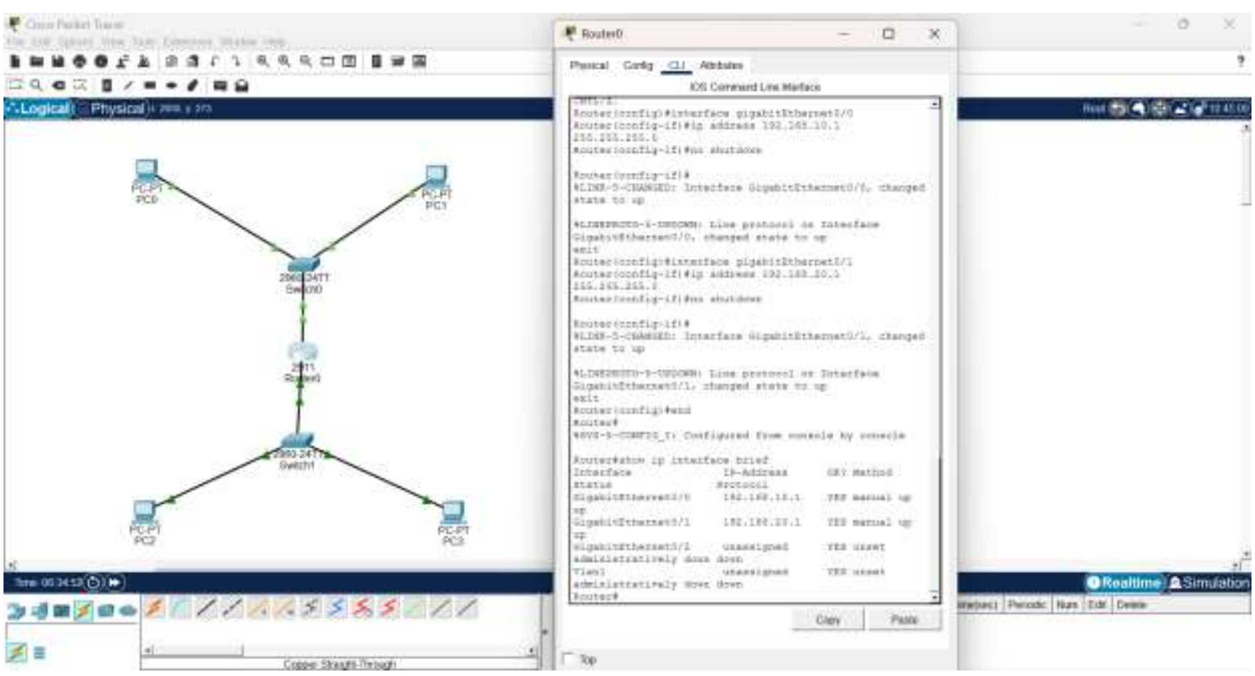
Configuration

Device	IP Address	Subnet Mask	Default Gateway
Router G0/0	192.168.10.1	255.255.255.0	—
Router G0/1	192.168.20.1	255.255.255.0	—
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.20.10	255.255.255.0	192.168.20.1
PC3	192.168.20.11	255.255.255.0	192.168.20.1

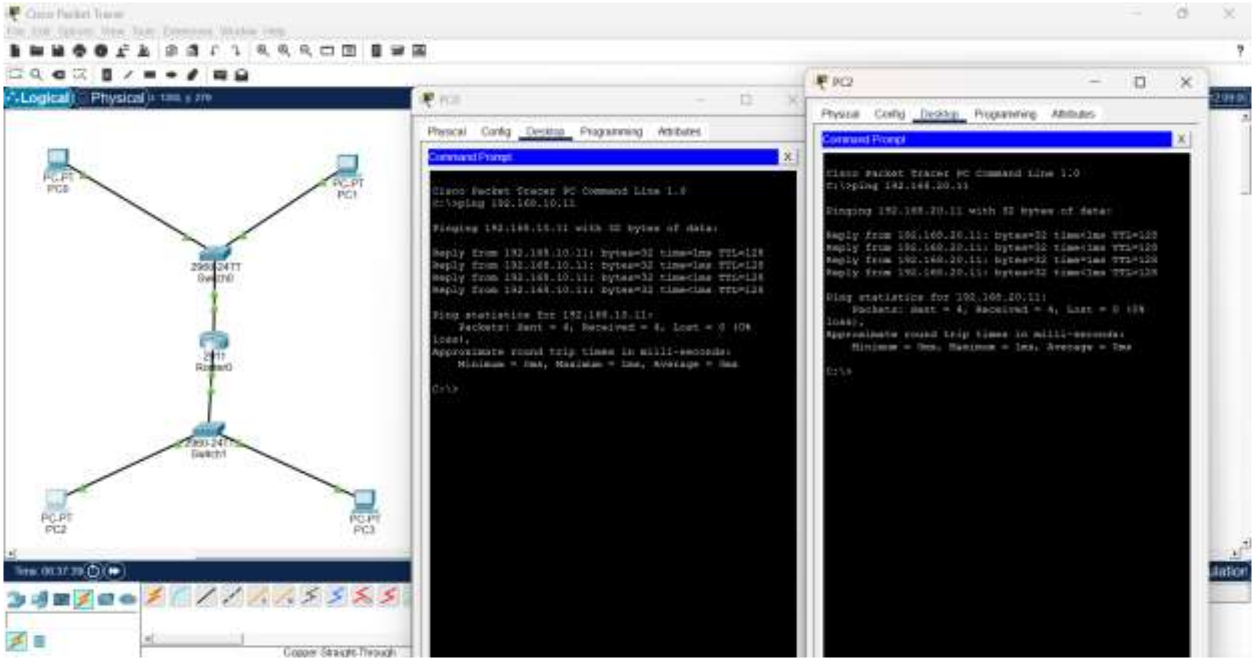
Explanation

The network consists of two LANs connected through Router0. LAN 1 uses the **192.168.10.0/24** network, with PC0 and PC1 using **192.168.10.10** and **192.168.10.11**, and Router G0/0 as the default gateway **192.168.10.1**. LAN 2 uses the **192.168.20.0/24** network, with PC2 and PC3 using **192.168.20.10** and **192.168.20.11**, and Router G0/1 as the default gateway **192.168.20.1**. The router forwards packets between the two different networks, enabling communication between all PCs.

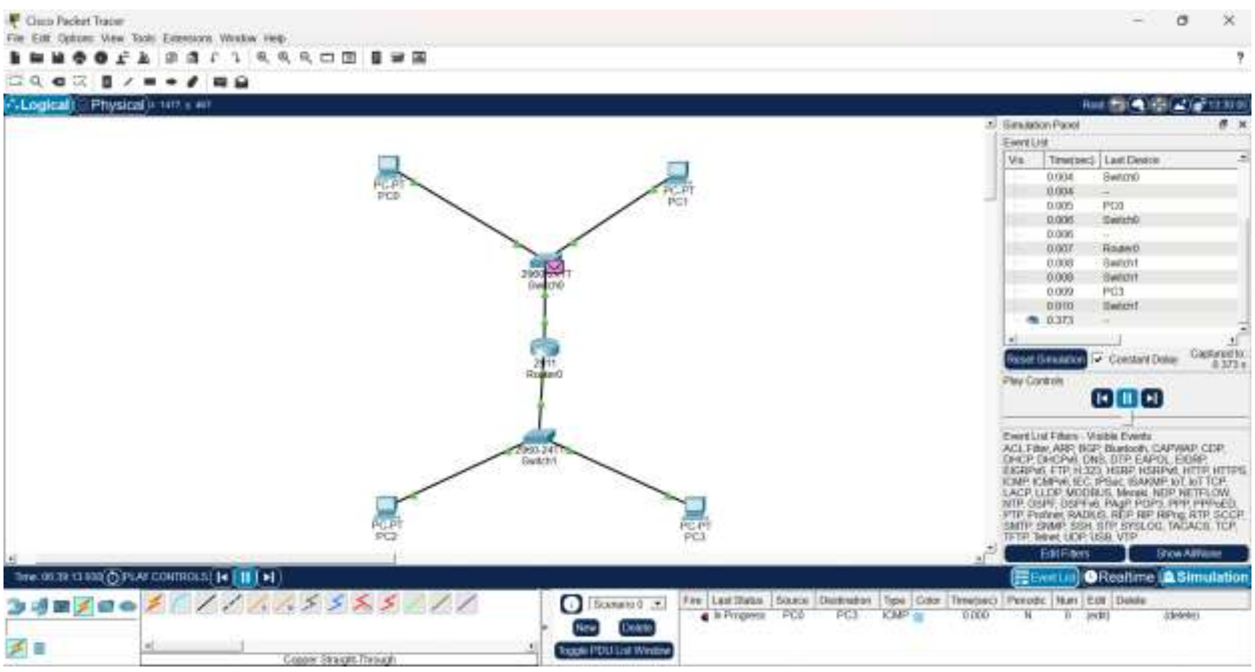
Router Configuration



Testing LAN 1 & LAN 2



Cisco Packet Tracer Simulation Output

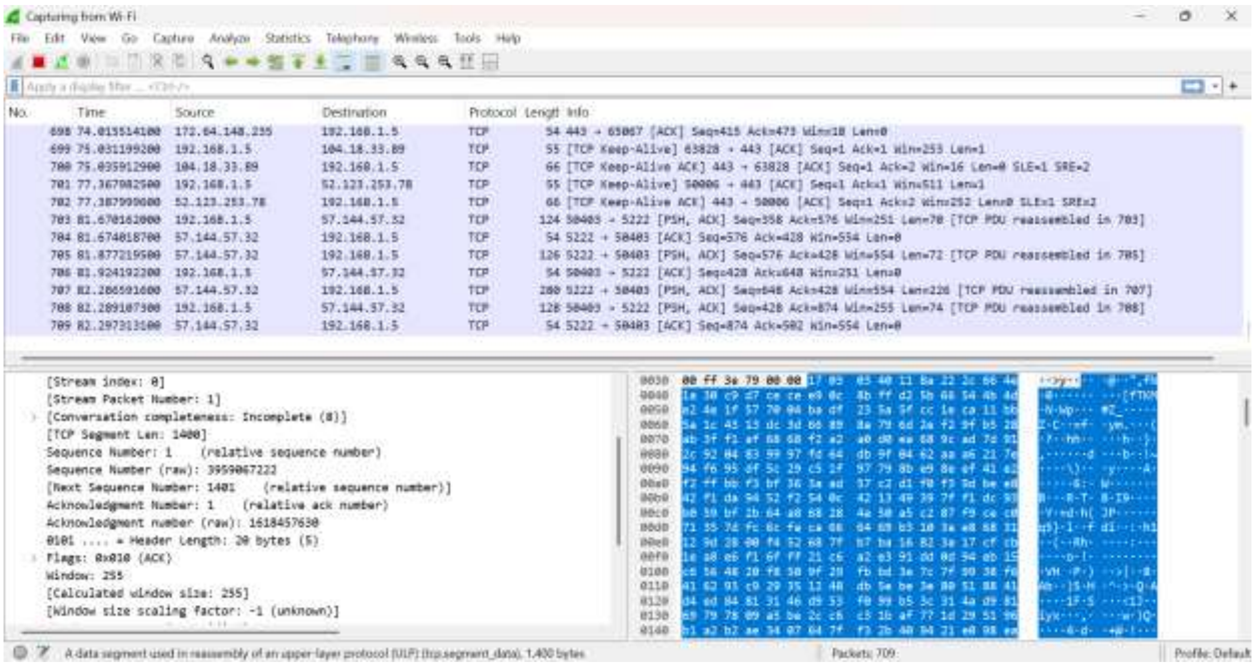


Observation

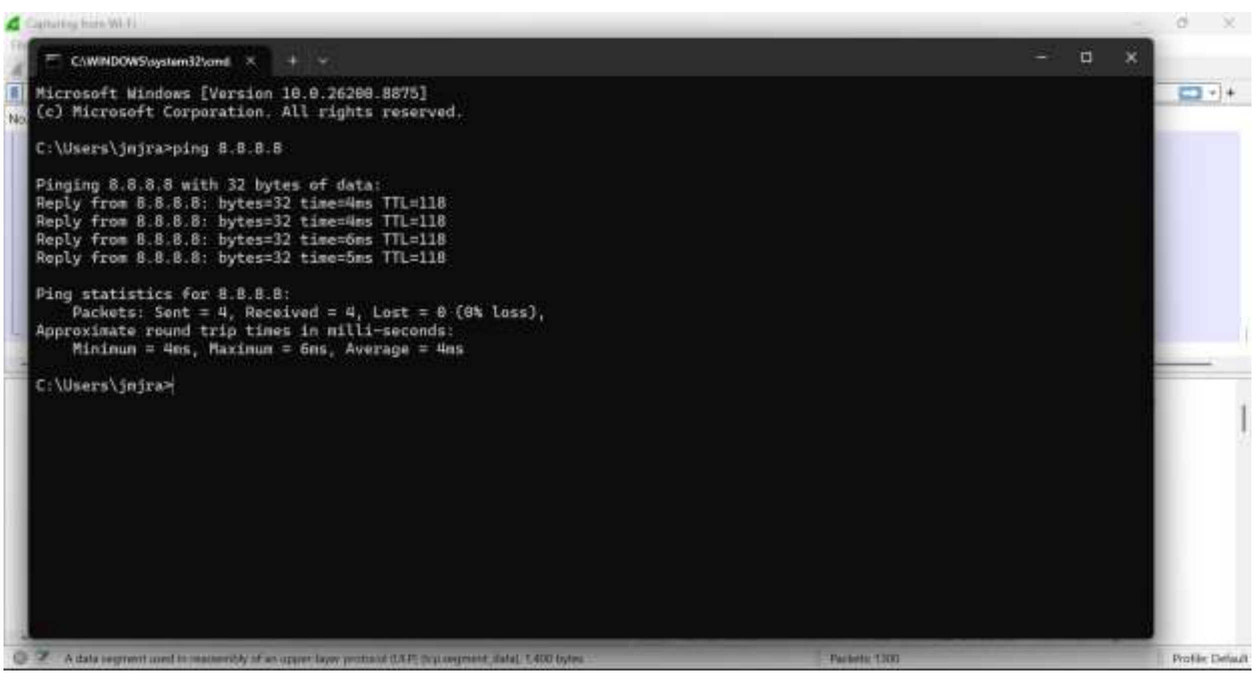
IPv4 Field	Observed Value
Source IP Address	192.168.10.10
Destination IP Address	192.168.20.10
TTL	Note the value shown in Simulation
Protocol Number	1
Header Length	20 bytes

Implementation

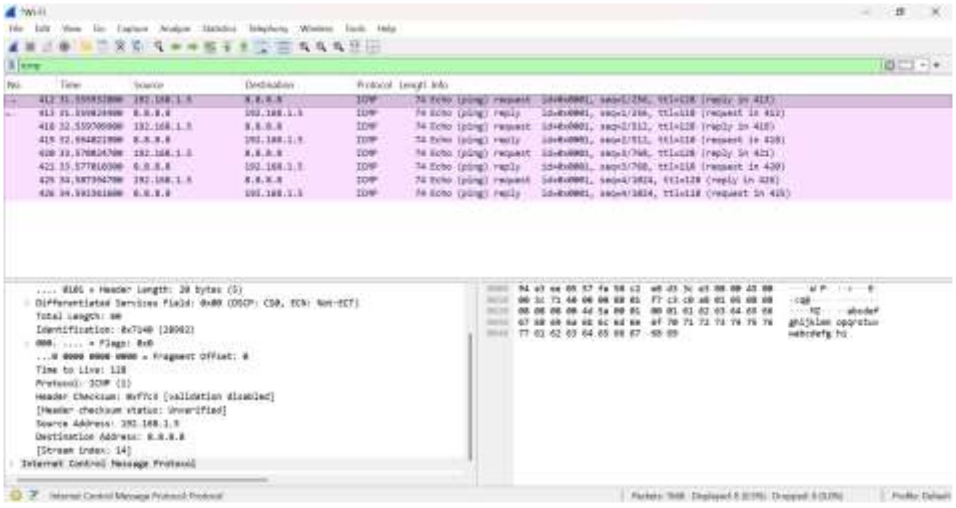
WireShark



Generating ICMP Packet



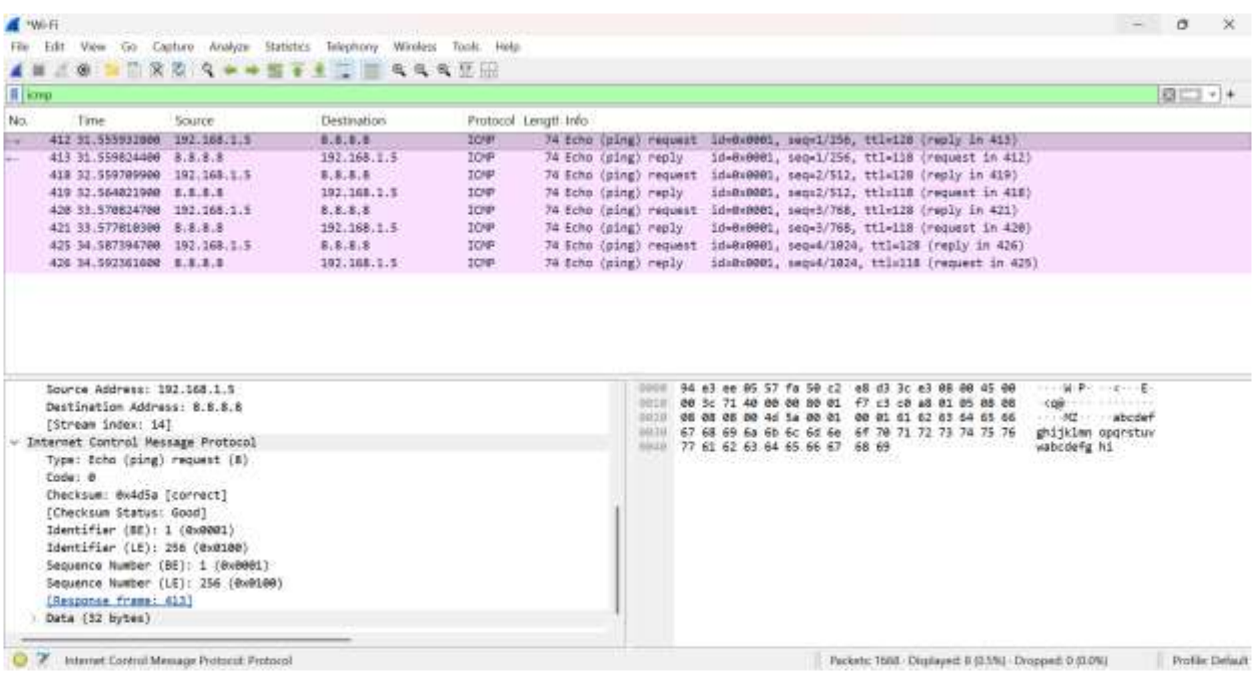
Finding Addressess



Observation

IPv4 Header Field	Value
Source IP Address	Your actual source IP
Destination IP Address	8.8.8.8
TTL	Value shown by Wireshark
Protocol Number	1
Header Length	20 bytes

Output



Explanation

The Wireshark capture successfully displayed the ICMP packets generated during the ping operation. The IPv4 header was analyzed to identify the Source IP Address, Destination IP Address, TTL, Protocol Number, and Header Length. The protocol number 1 confirmed that the packet uses ICMP, while the IPv4 header length was observed as 20 bytes. The capture demonstrated the transmission of ICMP Echo Request and Echo Reply packets between the source and destination.

Result

The two LANs were successfully connected through a router, and end-to-end communication was verified using **ping**. ICMP packets were successfully captured in Wireshark, and the **Source IP, Destination IP, TTL, Protocol Number, and Header Length** fields of the IPv4 header were analyzed. The role of the Network layer in forwarding packets between different networks was successfully demonstrated.

3. **Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.**

Aim

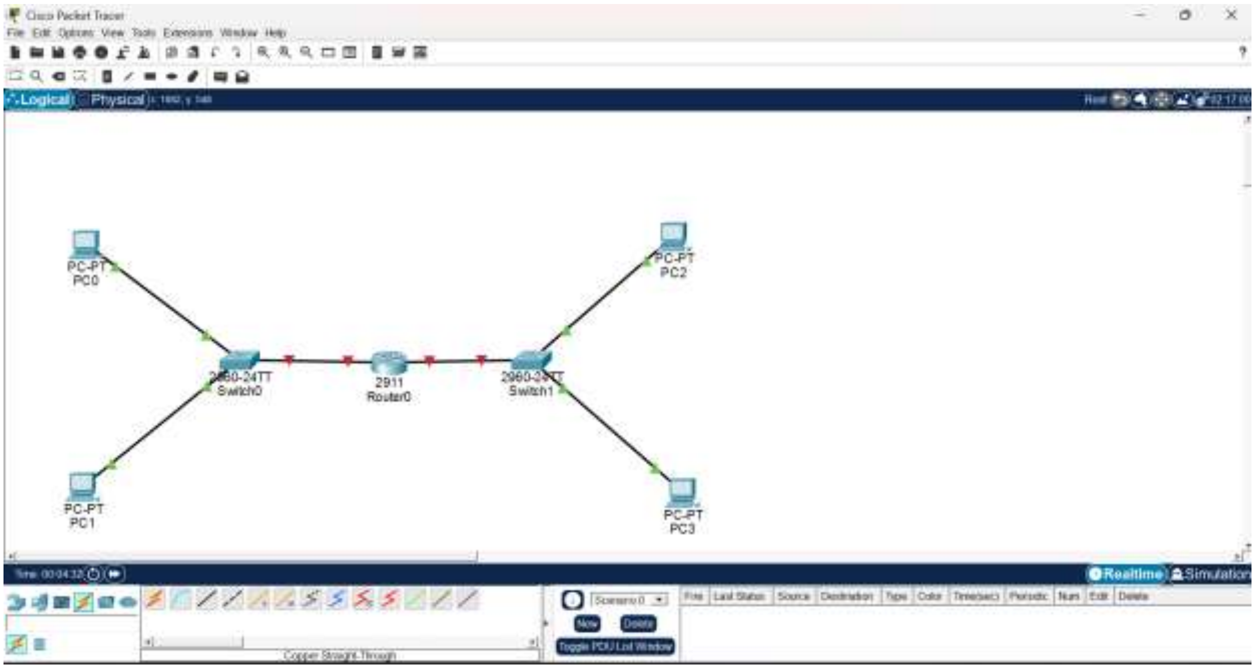
To design and configure two LANs connected through a router in Cisco Packet Tracer, verify end-to-end communication using ping, and analyze IPv4/ICMP packet header fields using Wireshark.

Procedure

1. Create two LANs using PCs and switches, and connect both LANs to a router.
2. Assign IPv4 addresses and subnet masks to all PCs and router interfaces.
3. Configure the default gateway of each PC according to its respective LAN.
4. Enable the router interfaces using the no shutdown command.
5. Verify connectivity between the PCs using the ping command.
6. Start Wireshark and capture the ICMP packets generated during the ping operation.
7. Analyze the IPv4 header fields such as Source IP, Destination IP, TTL, Protocol Number, and Header Length.
8. Observe the movement of ICMP packets through the router between the two different LANs.
9. Verify the role of the Network layer in forwarding packets between different networks.

Implementation

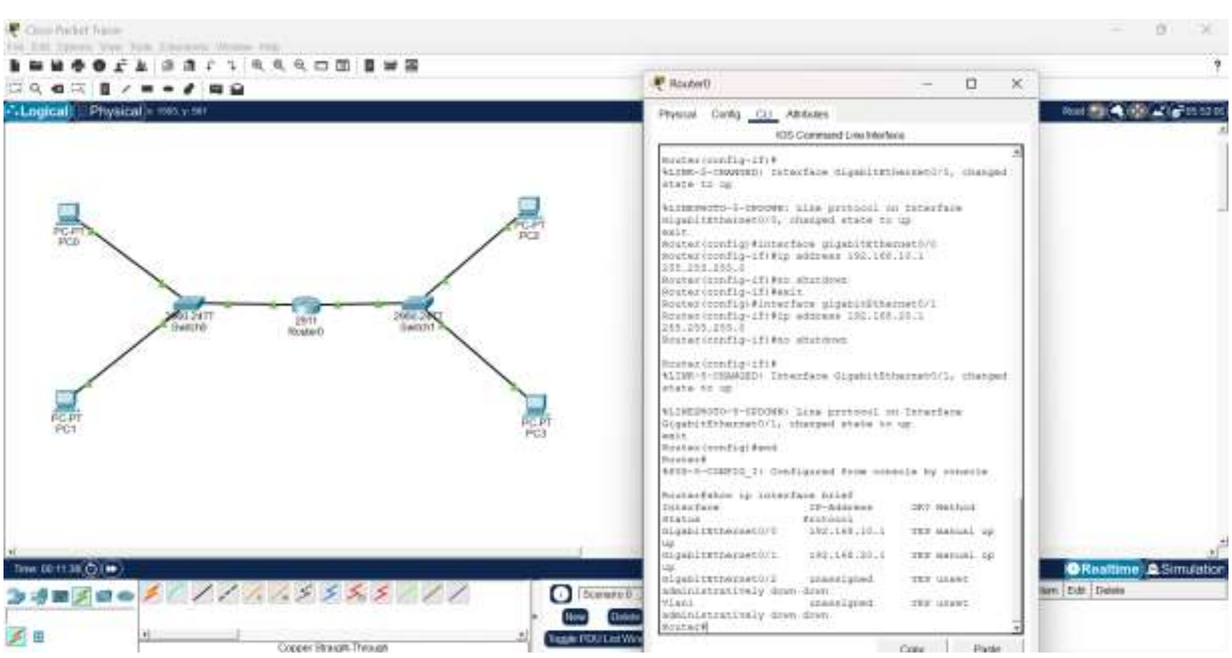
Cisco Packet Tracer



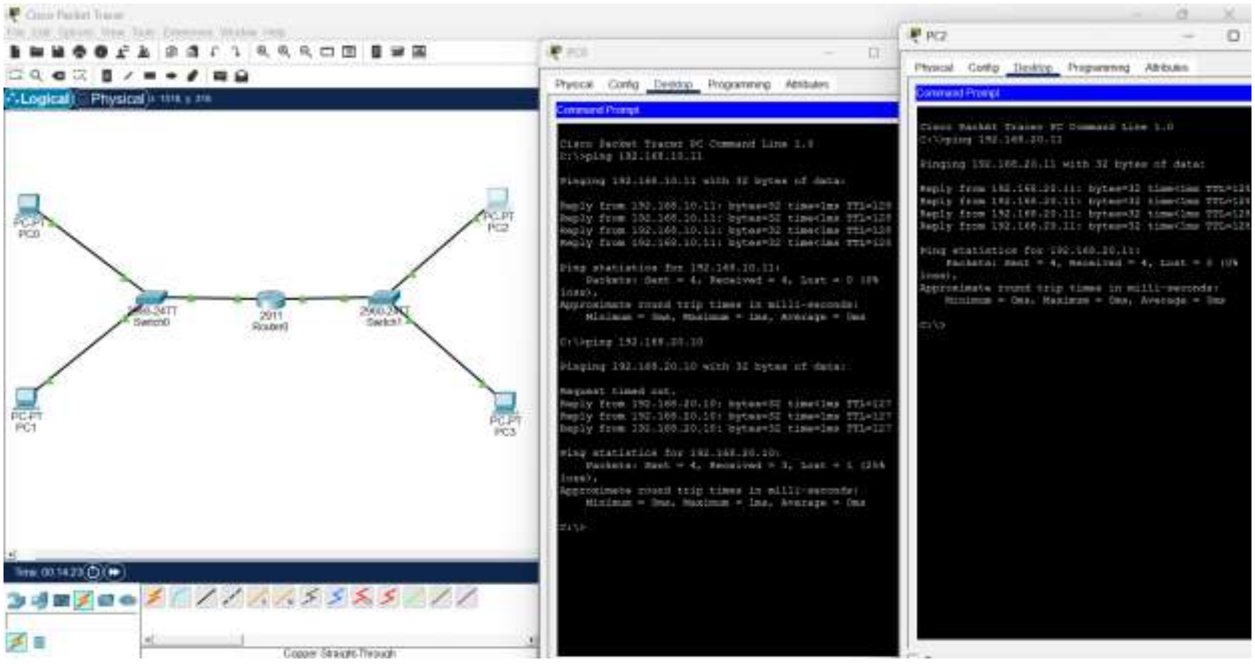
Configuration

Device	IP Address	Subnet Mask	Gateway
Router G0/0	192.168.10.1	255.255.255.0	—
Router G0/1	192.168.20.1	255.255.255.0	—
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.20.10	255.255.255.0	192.168.20.1
PC3	192.168.20.11	255.255.255.0	192.168.20.1

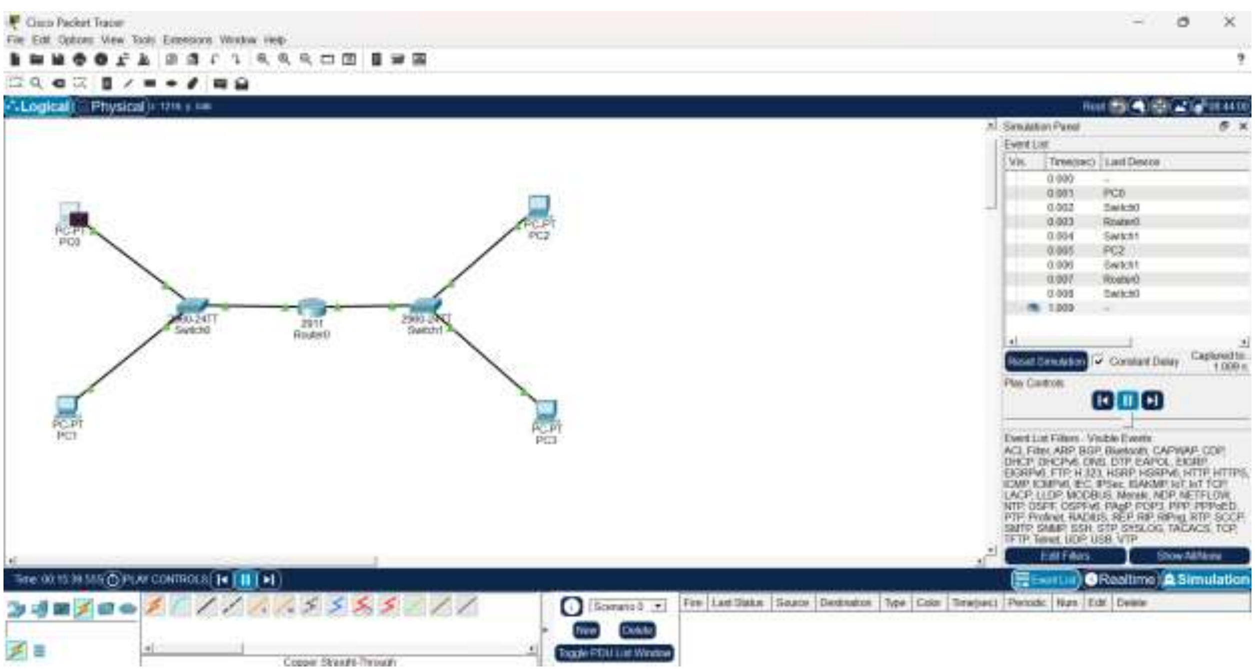
Router Configuration



Testing LAN 1 & LAN 2



Cisco Packet Tracer Simulation Output



Observation

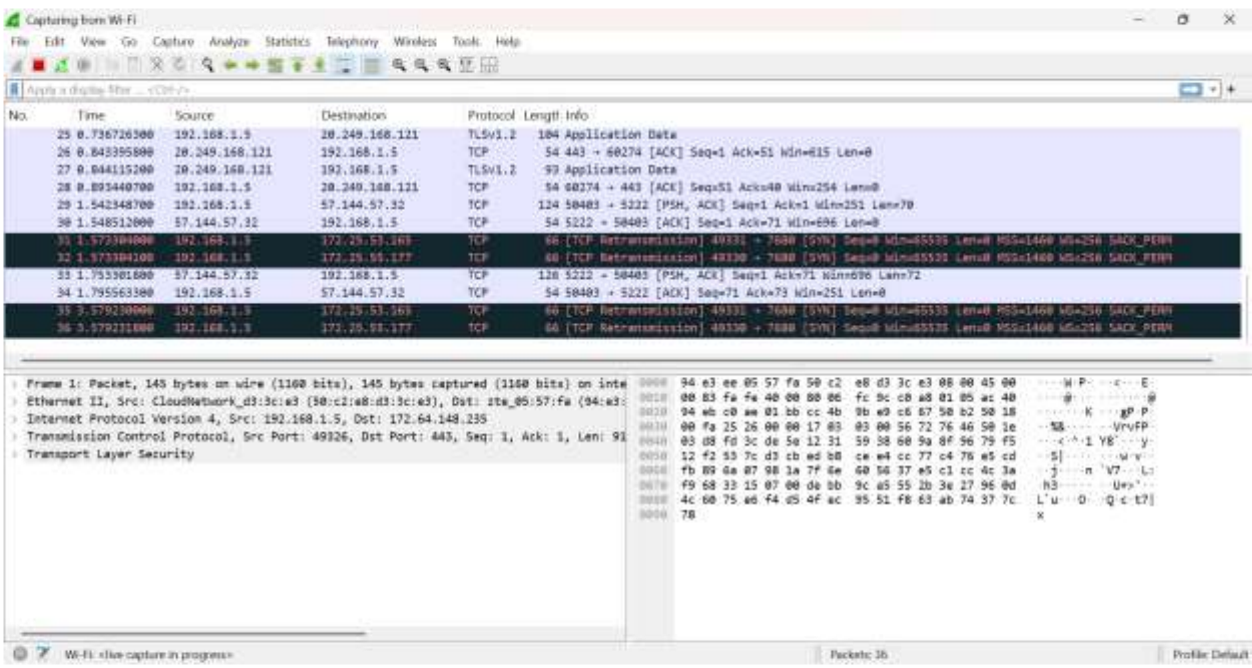
Field	Expected/Observed
Source IP	192.168.10.10
Destination IP	192.168.20.10
Protocol	ICMP (1)
TTL	Record the value shown
Header Length	Normally 20 bytes

Explanation

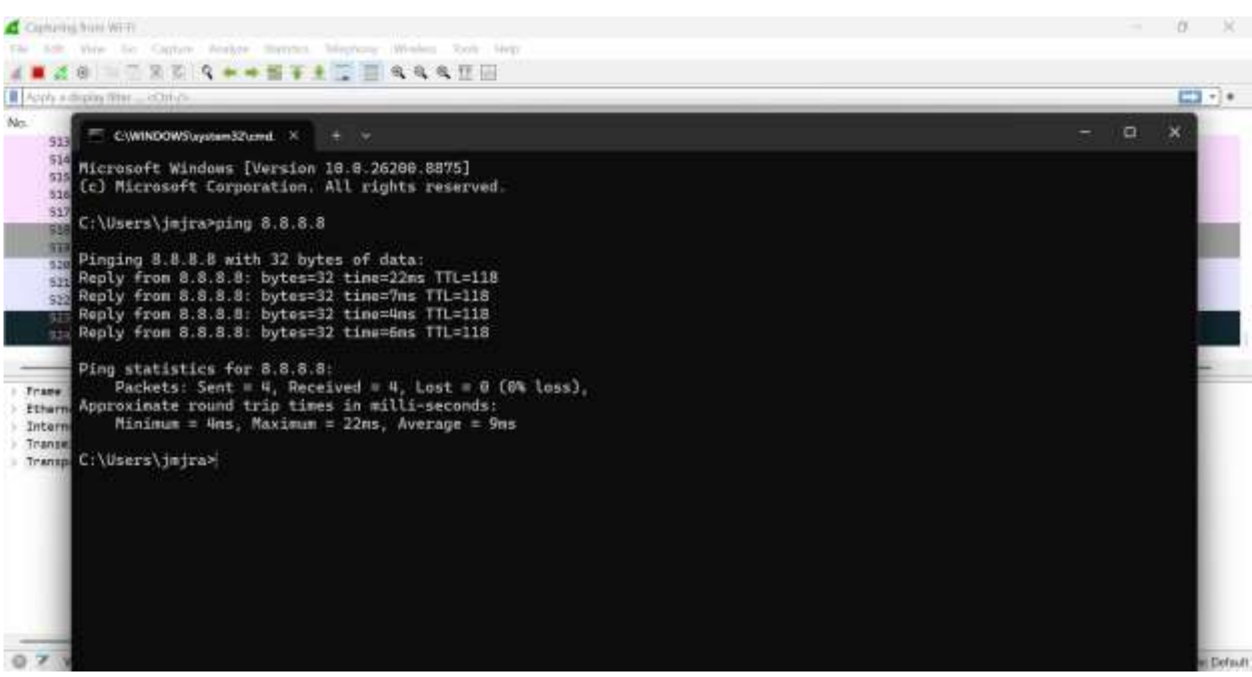
The two LANs were successfully connected through a router in Cisco Packet Tracer. IPv4 addresses, subnet masks, and default gateways were correctly configured, and successful **ping replies** confirmed communication between devices in both LANs. The router successfully forwarded ICMP packets between the **192.168.10.0/24** and **192.168.20.0/24** networks, demonstrating the role of the Network layer in inter-network communication.

Implementation

WireShark



Generating ICMP Packet



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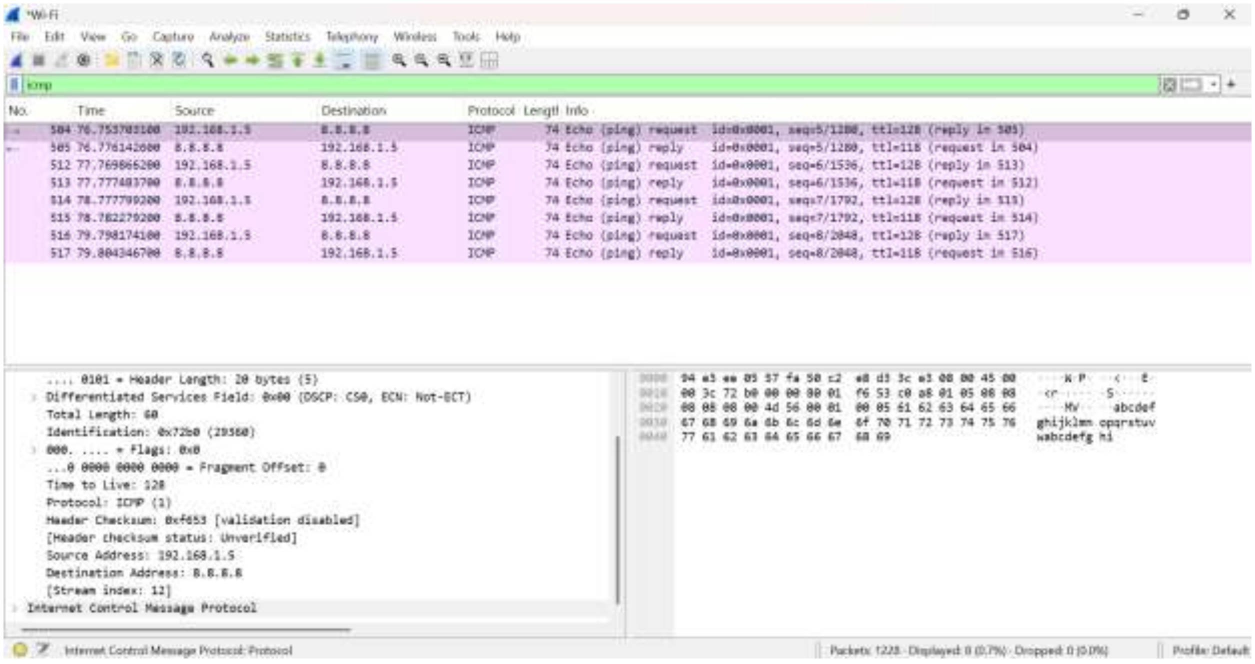
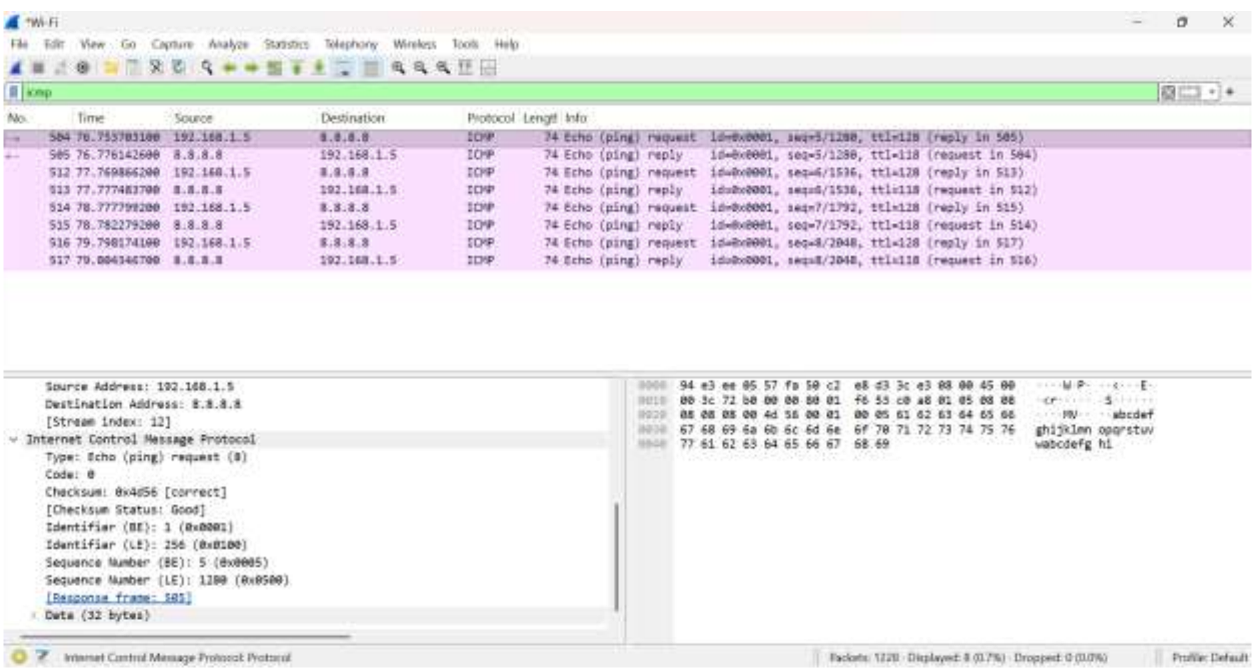


Output



Explanation

The Wireshark capture successfully identified the **ARP Request and ARP Reply** packets generated before communication. The ARP Request contained the sender's IP and MAC addresses and the target IP address, while the target MAC address was initially unknown. The ARP Reply provided the corresponding MAC address of the target device. This demonstrated how **ARP resolves an IP address into a MAC address** before actual data transmission takes place.

Result

The two LANs were successfully connected through a router, and end-to-end communication was verified using the ping command. ICMP packets were captured and the IPv4 header fields were analyzed, demonstrating how the Network layer forwards packets between different networks.